

Maximum position in a supercritical multi-type branching random walk with heavy-tailed displacements.

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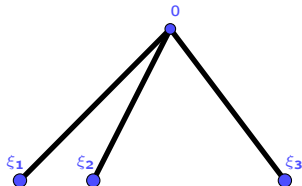
May 21, 2026

Branching random walk

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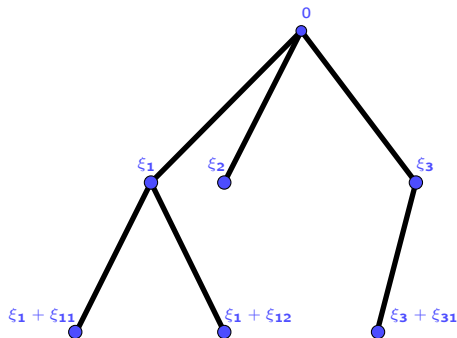


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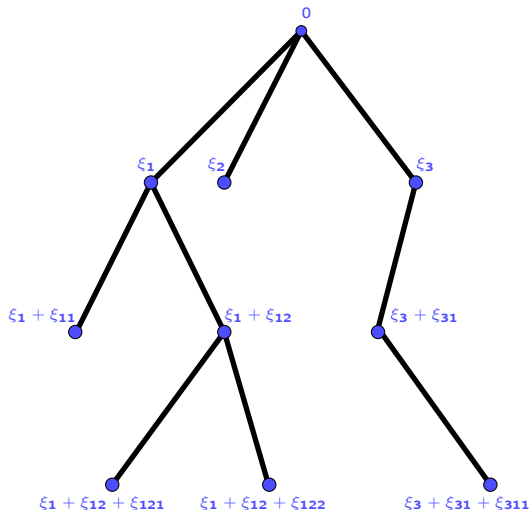
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$$v \in \mathbb{T}, \quad S_v = \sum_{i=1}^N \xi_{v_i}$$



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- Finite exponential moments (Biggins, 1976): Assume that for some $\theta > 0$,

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- Stretched exponential tails (Ganter, 2000): Assume that the displacements are i.i.d. and satisfy for some $r \in (0, 1)$,

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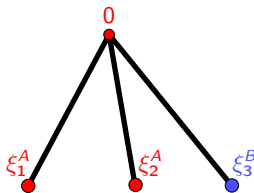
Multi-type branching random walk

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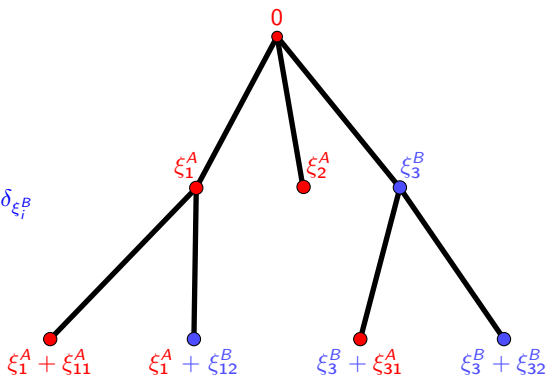
$$\mathcal{Z}_A = \mathcal{Z}_{AA} + \mathcal{Z}_{BB} = \sum_{i=1}^{N_{AA}} \delta_{\xi_i^A} + \sum_{i=1}^{N_{AB}} \delta_{\xi_i^B}$$



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$$Z_A^{(1)}, Z_A^{(2)}, Z_B^{(3)}, \dots$$



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The process (or the matrix M) is called **irreducible**, if any given type can appear in any line of descent, and **reducible** otherwise. In a two type model, the process is reducible when M is triangular.

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- Reducible case, deterministic model (Weinberger et al., 2002): Assume that M is reducible. Then R_n/n converges to a constant $\Gamma = \max\{\Gamma_A, \Gamma_B\}$, where Γ_i is the limiting constant in a model consisting only of type i .

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- Reducible case (Biggins, 2009): Assume that M is reducible. Then R_n/n converges almost surely to a constant, that can be strictly greater than the speed of either type considered in a one type model (**anomalous spreading**).

Theorem (K.K. 2025)

Assume that the displacements are independent and satisfy, for some $r_A, r_B \in (0, 1)$, $\mathbb{P}(\xi^i \geq t) = \exp\{-t^{r_i}\}$. Let $r = \min\{r_A, r_B\}$. Then

$$\frac{R_n}{n^{1/r}} \xrightarrow{\text{a.s.}} (\log \rho)^{\frac{1}{r}},$$

where:

- When M is irreducible, ρ is the largest eigenvalue of M .
- When M is reducible, then

$$\rho = \begin{cases} \max\{\mathbb{E}[N_{AA}], \mathbb{E}[N_{BB}]\} & r_A \geq r_B, \\ \mathbb{E}[N_{AA}] & r_A < r_B \end{cases}$$

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- The maximum of K random variables with stretched exponential tails grows as $(\log K)^{1/r} = (n \log \rho_i)^{1/r_i}$ when $K \sim \rho_i^n$ and $r = r_i$, so heaviest tail will always win.

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Theorem (Bhattacharya, Maulik, Palmowski, Roy 2019, K.K. 2025)

Assume that the displacements are independent and satisfy

$$\mathbb{P}(\xi^i > x) \sim x^{-r_i} \quad \text{as } x \rightarrow \infty$$

for some positive constants $\{r_i\}_{i \in \{A,B\}}$, and that M is irreducible. Then

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converges in law, where ρ is the principal eigenvalue of M , and $r = \min\{r_A, r_B\}$.

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- Proof: apply the principle of a single big jump, note that the asymptotic number of particles of either type is ρ^n , get the result.

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- The heaviest tail does not necessarily win. Since the maximum of K independent random variables with regularly varying tails grows roughly as $K^{1/r}$, to find the correct normalization we need to compare the quantities ρ_i^{1/r_i} .

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- If $\mathbb{E}[N_{AA}] = \mathbb{E}[N_{BB}]$ and $r_A \geq r_B$, $\gamma_n = (n\mathbb{E}[N_{BB}])^{1/r_B}$.

- The heaviest tail does not necessarily win. Since the maximum of K independent random variables with regularly varying tails grows roughly as $K^{1/r}$, to find the correct normalization we need to compare the quantities ρ_i^{1/r_i} .
- When the matrix is reducible, the relevant eigenvalues may not be simple. The asymptotic number of type B particles is then $n\mathbb{E}[N_{AA}]^n$.

Based on:

Krzysztof Kowalski, (2025) *Asymptotic behavior of the extremal position in a multi-type branching random walk with heavy-tailed displacements*, arXiv:2509.14808